NVIDIA Geforce GTX 980

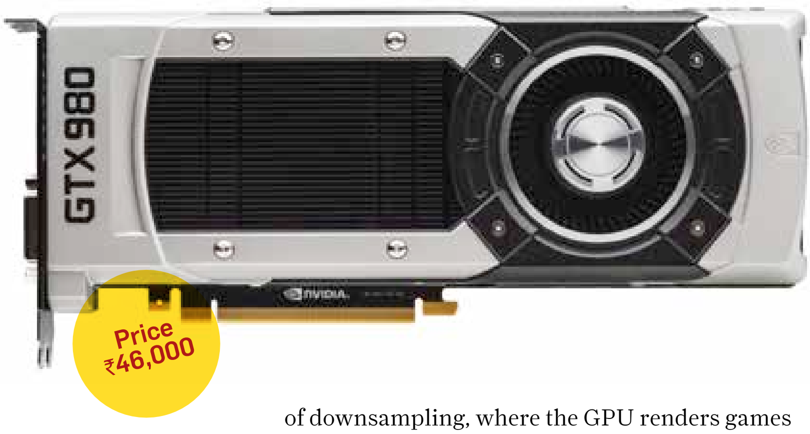
The fully fledged Maxwell GPU packs a wallop

Maxwell debuted with the GTX 750 Ti and it brought with it quite a few changes. The architecture as a whole was reorganised to make the Streaming Multiprocessors (SM) much more efficient to the extent that lesser number of transistors have resulted in performance similar to that of Kepler. But the GTX 980 is based on Maxwell's second generation and has more features than the 750 Ti and a lot more than Kepler. Here's a brief roundup of the new features.

So what's new with the features maxwell offers on the 980?

Dynamic Super Resolution (DSR)

If you are familiar with the process of supersampling (used in spatial anti-aliasing), or the method



of downsampling, where the GPU renders games at higher resolutions, and then scales the image down to the monitor's native resolution, DSR should be easy to understand because essentially it's the same thing.

What used to be a hack (dgit.in/1C4d89t) earlier, thanks to DSR – perhaps for the first time ever – comes as a feature built into the driver itself. It lets gamers who don't have 4K or UHD displays (yet) to utilize the extra horsepower of their GPUs to up the gaming experience on their 1080p monitors. Unlike downsampling which sometimes can generate artifacts on textures when certain post-processing effects are applied, Dynamic Super Resolution uses a 13-tap Gaussian filter during the conversion process (downscaling) and is supposed to reduce or eliminate the aliasing artifacts experienced with plain downsampling, which relies on a simpler box filter.

The GTX 980 whitepaper however says, "While it's compatible with all GeForce GPUs, the best performance can be seen when using a GeForce GTX 980."

MFAA

The next big piece of technology announced was closely related to DSR. While DSR tries to optimise games for scaling in the visual domain MFAA tries to better the performance aspect. In a nutshell, it gives the user the same experience as 4X MSAA at the compute cost of 2X MSAA. How does it achieve that? Well that's a little complicated. Put on your Portal face and read on.

MFAA stands for Multi Frame-sampled Anti Aliasing and is a flexible sampling method available to Maxwell GPUs that tries to push AA quality and efficiency up a notch. MFAA alternates AA sample patterns both temporally and spatially to produce the best image quality.

What MFAA does is it algorithmically flips or mirrors samples of pixel coverage and combines by applying a temporal synthesis filter to create a coverage pattern identical to 4x MFAA. This saves on hardware compute cost.

VXGI

NVIDIA is confident Voxel Global Illumination is a huge step towards the holy grail of real time computer graphics – path tracing. It is known that rendered scenes start to look closer and closer to reality when GPUs begin to process those millions of light rays, millions of bounces and resultant energy losses in the intensity of light in a scene – all in real time. A tall order indeed. To circumvent this game designers use many techniques such as pre-baking textures or ambient occlusion (commonly referred to as the poor man's global illumination). With Voxelization NVIDIA hopes to bring in game lighting one more step closer to this dream of "true" path tracing.

This is how VXGI works:

Begin by converting an entire scene into voxels (three dimensional pixel). These voxels store two types of information – how much of the voxel is occupied by the actual object and the intensity and direction of light being bounced off that part of the object. This light that is bouncing off is calculated using a new method of tracing called cone tracing. It's more like an approximation or "big picture" approach to replicating diffused light. But it does